

Sovereign Agentic OS

The governed, EU-sovereign operating system for data, knowledge, agents and software — where AI gets real, safe hands on your work.

Orchestrated by Data Masterclass • datamasterclass.com • www.sovereign-agentic.com

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1 The one-paragraph version

The **Sovereign Agentic OS** is a self-hostable, EU-residency operating system for your data, knowledge, agents and software – a single **governed** stack where every action, taken by a person *or* by an AI, runs **as you**: OPA-policy-checked, row- and column-secured, and Langfuse-audited. It assembles roughly two dozen best-in-class, permissively-licensed open-source tools – a Trino/Iceberg lakehouse, a Cube semantic layer, OpenSearch retrieval, a LiteLLM model gateway to sovereign EU models, LangGraph agents – into one operating model you learn once and apply everywhere. Its promise is simple and, we think, important: it gives AI **real, safe hands on your data**. The web UI and the AI (over MCP) travel the *exact same governed path* – there is no privileged back door – so you can hand an agent the keys to your lakehouse and know, provably, that it can never do anything you couldn't do yourself.

Where to go next. Curious what it feels like? Read *The guided tour*. Want to run it? Jump to *Quickstart*. Coming from a governance or security background? Start with *The governance model*. Contributing code? *How to contribute* and `os-ui/ARCHITECTURE.md` are your on-ramp.

2 Why this exists

For two years the industry has been stuck on the same tension. **Agents are only useful when they can act** – read the warehouse, write a table, ship an app, call a tool. But the moment an agent can act, the honest questions arrive: *Whose identity is it running as? What is it allowed to touch? Where did the data go? Can I prove, after the fact, exactly what it did?* Most stacks answer these by wrapping an autonomous agent in ad-hoc guardrails bolted on after the fact – and by shipping your prompts and data to a US-hosted model.

The Sovereign Agentic OS answers them structurally, with three ideas doing all the work:

1. **One governed path for humans and AI.** Every write – from a button click, from a tab's built-in assistant, from Claude or ChatGPT over MCP, or from an autonomous agent – flows through the *same* function: `authorize → act → trace`. Governance isn't a feature you can forget to turn on; it's the only road in.
2. **Sovereignty is the substrate, not a badge.** The production target is **STACKIT** Kubernetes with STACKIT Object Storage in **EU01 / Deutschland Süd**, and model calls route to **STACKIT AI Model Serving**, so prompts and completions never leave the EU boundary. Default-deny egress means an agent has *no* raw internet unless you grant it.
3. **Permissive open source, end to end.** Every bundled component is Apache-2.0 / MIT / BSD / ISC-class licensed – full auditability, no proprietary lock-in, the right to host and modify it forever. The core is **Apache-2.0**, and the permissive posture is *enforced*, not merely promised: an automated `check:licenses` gate (`license-checker --onlyAllow`) rejects any dependency outside a strictly permissive allowlist (MIT, Apache-2.0, ISC, BSD-2/3-Clause, OBSD, and a few equivalents) – so no copyleft can slip in – with a CycloneDX SBOM (`sbom.cdx.json`) and a full attribution manifest (`THIRD-PARTY-LICENSES.md`) alongside.

2.1 Who it's for

- **Regulated organizations, the public sector, and EU enterprises** that need data residency, a complete audit trail, and zero dependency on US-controlled cloud or hosted LLMs – but still want production-grade agentic workflows.
- **Data and platform leaders** who want *one* place for the lakehouse, the semantic layer, the knowledge spine, BI, and software delivery – instead of a dozen disconnected tools and a governance story stitched across all of them.
- **Curious engineers and OSS contributors** who want to see how a governed agent runtime is actually built – and to extend it, one clean tab-module at a time.
- **Data Masterclass participants** building real agentic systems on the *same* production components used in the field, not a teaching fork.

2.2 The problems it solves

THE TENSION	HOW THE OS RESOLVES IT
Governance vs. agent autonomy	Agents are genuinely autonomous <i>inside</i> a policy envelope. Every tool call is OPA-authorized; anything out of scope doesn't fail silently – it queues to a human as an approval card.
Sovereignty vs. capability	The full stack runs in your EU region on open source. Models are sovereign (STACKIT) yet swappable; nothing phones home.
Ten tools vs. one source of truth	One lakehouse, one semantic layer, one knowledge spine, one identity, one audit – so BI, agents and dashboards can never disagree about "revenue."
Demos vs. production	The same Helm chart runs on a laptop (<code>kind</code>) and on a sovereign STACKIT cluster. The only difference is a values choice per backend.

3 The operating model — learn it once

One idea ties every screen together. Internalize it here and the whole product reads the same.

3.1 Everything is a governed artifact

Whatever you make — a dataset, a knowledge workflow, a file, an agent, an app, an ML model, a metric, a connection, a dashboard — is an **artifact** with the same four attributes: **owner • domain • type • visibility**. Whatever its type, it travels one lifecycle:

Create → Document → Use → Promote — authored in the UI (which scaffolds the *real* tool underneath: a dbt model, a metric's semantic declaration, a Forgejo repo, a KServe service), preview-first, cataloged and audited.

3.2 One artifact model — every build tab reads the same

Every artifact in the OS reads the same way: a tab lists tiles in folders; **New** opens a short **type chooser** and lands you in **Edit**; clicking a tile opens a full-page **View**; a  **Edit** button re-opens it. Learn one tab and you can drive them all, because they share the same shape:

- **New → type → Edit → View.** The Context tabs (Data, Metrics, Files, Knowledge, Connections) and Dashboards all speak this one language — no staged steppers. Edit is a set of **named sections, each with its own Save** (e.g. Data's *Ingestion • Documentation • Transformation • Checks*), so you save the part you changed and nothing else. **View** is the read-first surface (talk to it, preview it, read its stats and quality). The build tabs that produce *running systems* — **Agents** (Define • Design • Build • Run • Evaluate), **Software** (Define • Design • Build • Test • Publish) and **Science** (Define • Train • Deploy • Predict • Monitor) — keep a **staged builder** (`lib/core/stages.ts` + `components/core/StageShell.tsx`), because a deploy genuinely has ordered gates; the artifact tabs no longer do.
- **Honest state.** Nothing shows a ✓ it hasn't earned: a data-quality scorecard turns green only when checks actually ran and passed (honestly `unknown`, never a fake pass, when nothing ran), a metric previews the real governed query, and a dashboard panel shows a live / offline-mock badge for whether Cube answered. No faked green.
- **AI where the work happens.** The Sovereign-OS AI helper is scoped to what you're doing and acts through the same governed tools. On the Data tab it dissolves into the flow — each section carries big ✨ **actions** ("✨ Draft documentation", "✨ Clean it up", "✨ Suggest measures" ...) — the same governed, audited, cost-capped assistant underneath, surfaced exactly where the work happens.
- **Simple ⇌ Developer, and lifecycle-in-header.** A **Simple** view keeps the flow calm and NL-first; a **Developer** view exposes the raw technical surface (the dbt SQL, the Cube YAML, the lineage graph, the repo tree). The artifact's name, visibility badge and lifecycle controls (Archive •

Restore • Delete • Promote) live in a **persistent header**, always reachable — this is where a dataset, metric or dashboard is **shared**, not a separate stage.

3.3 The sharing ladder

Visibility widens **one rung at a time**, and each move is strictly **two-step** — the person who *triggers* a promotion is never the person who *approves* it:

The OS speaks **one scope vocabulary everywhere** — **My • Domain • Company** (defined once in core, `lib/core/scopes.ts`). "My" is your private draft space; "Domain" is your team; "Company" is the whole tenant. Promote reads "**Promote to Domain**" and certify "**...to Company**" — the same two verbs on every tab.

Your "My" work is yours — no approval, ever. A builder — *and the agents they build* — create and write their own **personal** artifacts of every type (data, files, knowledge, metrics, connections, software, dashboards, agents, science) **directly**, with no admin review. The person's own rights and ownership are the authority. Approval only appears the moment you widen the audience:

SCOPE	MEANING	WHO TRIGGERS	WHO APPROVES
My	you only — the default; full rights, no approval	—	—
Domain	usable across the owning domain	the owner files a promotion request	a domain admin of that domain
Company <i>(certified)</i>	discoverable and importable by <i>other</i> domains, listed in the Marketplace storefront	a Builder / Domain admin — the domain vouches for it	the tenant Administrator — the platform accepts it

You can only change what you originally built. Edit rights follow ownership, not tier — even a shared artifact is editable only by its original owner (and, for Domain/Company artifacts, an in-scope admin); a personal "My" artifact is strictly owner-only, invisible to admins.

Approving is the action. High-stakes actions never fail silently: when you promote, certify or request a deploy you get one calm confirmation — *"Request filed — awaiting approval to Domain/Company"* — with a **Go to Policies & Approvals** → button that deep-links to and highlights the exact request. An admin who *can* approve it sees an **Approve now** button that approves inline (fail-closed — non-approvers never see it, and the server re-checks). On approve, the platform executes the governed effect — for a dataset, a physical publish; the tier flips only once it verifies — and writes the audit. Nothing enters the governed store without **documentation + passing checks**: a transparency gate that turns green only when an artifact is documented and in the lineage graph.

The ladder also runs down — on every artifact type. From an artifact's header, **Unshare** returns a Domain artifact to My (its owner or an in-domain Domain admin) and **Revoke from Company** returns a certified one to Domain (Admin). A metric shares its dataset's tier, so demoting a metric moves the dataset and every metric on it together — the confirm dialog says so.

3.4 Four roles, assigned per domain

The ladder is exactly **creator < builder < domain_admin < admin**.

ROLE	WHAT THEY DO
Creator	the base role – creates and runs their own artifacts (My scope by default) and consumes anything at Domain or Company scope. Files promotion requests; cannot approve.
Builder	the domain approver – everything a Creator can, plus review/approve My→Domain promotions, deploys, knowledge and connections. An approver, <i>not</i> a people-admin.
Domain admin	everything a Builder can, plus administering the users of their own domain(s) only – invite, edit, assign roles up to Builder . Never mints another Domain admin.
Administrator	tenant-wide – the only role that appoints Domain admins ; sets policy, certifies to Company scope (the Marketplace storefront), sets cost caps; runs the Admin section.

Roles are assigned **per domain** and **compiled to OPA**, so a person who is a Builder in one domain and a Creator in another sees exactly the right controls in every tab, instantly.

3.5 Every assistant acts, and acts as *you*

Each tab has a built-in assistant, and there is one **overarching assistant** – **Ask the OS** – present on every tab. Neither is a chat box. Each runs one loop: **PLAN** with the reasoning model, then **ACT** in a bounded tool-calling loop with the worker model – calling that tab's governed tools, which are the *same* OPA-authorized, Langfuse-traced functions the UI uses. Crucially, **Ask the OS is a client of the OS's own MCP server**: it dispatches through the exact same governed path (`handleRpc`) that Claude Desktop uses, under *your* delegated identity. It inherits identical guardrails – role floor, approval gates, OPA/RLS, audit – and it is transparent: each answer lists the governed tools it invoked, and if governance *blocks* a call it says so plainly. There is **no privileged side-channel**.

4 The guided tour

Open the OS in a browser and you land on a left sidebar organized into named sections. Here is the whole map, in sidebar order – each surface in a few vivid sentences, with what you *do* there.

4.1 Entry

- **Home – the golden-path launcher.** The warm front door after you pick a domain. An illustrated launcher of the golden paths (Data, Knowledge, Agents, Software, Science, Metrics, Dashboards, Big Bets, Marketplace, Connections), each card with a role-aware action. It *only* orients and routes – the live view lives one click away in Cockpit.
- **Cockpit – what's moving, what needs you.** A persona-ordered live overview: a pulse strip (*Needs you • In progress • Your items • Spend vs. cap*), your work-in-progress, and a scannable "top items, by type" board. Cockpit *reads and routes* – it never recomputes another tab's numbers and never bypasses governance.
- **Marketplace – consume across domains.** The *Consume* counterpart to every tab's *certify to Company* step: discover and reuse Administrator-certified (Company-scope) products of every type across the tenant. The storefront keeps its name; importing is a **governed grant**, not a copy – the default is read-in-place, under your own identity and row-level security.

4.2 Plan

- **Strategy – pillars, value and adoption.** Where the company plans its agentic transformation, in exactly three calm sections: **Strategic Pillars** (each with a big value target; Big Bets are created under them), **Self Service** (how broadly your people build for themselves), and **Foundations** (the certified asset base every bet builds on).
- **Big Bets – initiative roadmaps.** A strategic AI bet as a **goal + dated roadmap** built from real artifacts across the platform. **Every bet sits under a Strategic Pillar** – creation requires one from both entry points (the Big Bets "New" panel and "New bet under this pillar" from Strategy) and via MCP. Status derives *live* from each artifact's real lifecycle; the roadmap rolls up on-track / at-risk.
- **Operating Model – how the company runs, at three scopes.** *My / Domain / Company Operating Model*, each a fixed set of sections – **General • Strategy • Business • Organization • Architecture • Data • Glossary** – governed per scope (My = owner, Domain = domain_admin+, Company = admin). It's the durable, structured backbone agents can be granted as context.
- **Workflows – the process spine.** A **workflow** per business process (ordered steps • business rules • expert knowledge, each step owned by a Human / Software / Agent / external actor), retrievable and grantable to agents – with a **Data & Metrics** tab that links the governed datasets and KPIs the process runs on, each a scope-badged chip deep-linking to the real artifact.
- **MCP (Builder+)** – the setup surface for connecting external AI clients over MCP.

- **Tutorials.** An illustrated, hands-on tutorial for every tab – fourteen today, each kept in step with its tab's current journey – reached from Home or a tab header, that can spotlight the real controls and let you practice in a sandbox before doing it for real.

4.3 Context

- **Knowledge – the domain's captured know-how.** Human-authored reference knowledge, made retrievable by a knowledge agent behind document-level security. Mark a decision rule **hard** and it compiles into an OPA guardrail. (The structured backbone – Operating Model, Strategy, Big Bets, Workflows – lives in the Plan section; Knowledge is the reference library.)
- **Files – a calm governed drive.** Any unstructured file – documents, images, audio, video – added via **New (Upload a file or New note (markdown))** written in place) and auto-indexed (parse → embed → hybrid OpenSearch) so agents can search and cite it. Opening a file lands on a **full-page reading surface**. Governed exactly like Data; *"Use as"* distils a file into Knowledge or Data.
- **Data – datasets, refined and governed.** One artifact per dataset, two surfaces – **Edit** and **View** – that turn a plain-language flow into real governed artifacts (a dlt pipeline, dbt models, a Cube cube) with no YAML. **New** opens a calm two-path chooser first: 📁 **Ingest new data** (bring a file or extract in) or 🔗 **Create a curated dataset** (combine governed datasets you can read into one new joined table). Either lands you in **Edit**; a name already taken in the domain isn't a dead end – an inline note explains the clash and offers a one-click **Open** of the existing dataset, or a distinguishing name – and renaming later is a labelled 🏷️ **Rename** button (the physical table slug stays stable). Tiles group into **Ingested Data** and **Curated Data**. **Edit** is a set of named sections, each with its own Save – no stepper, no Continue: an **ingested** dataset walks **Ingestion** → **Documentation** → **Transformation** → **Checks** (*Save Data • Save Documentation • Save Transformations • Save Data Quality Checks*); a **curated** dataset opens on **Composition** (pick an explicit **base dataset**, add **joins**, keep or rename columns, add **derived fields**, then *Save Composition*) followed by Documentation and Checks. Ingested data **never joins** – that is the curated path. The business (Gold) layer that metrics read is **materialized automatically**: a clean single-table ingested dataset **passes through** to a queryable business table with no manual "build Gold" step, and a curated dataset materializes the composition you saved. Measures are defined in the Metrics tab, not in the composition, so the business layer stays a pure projection/join and a rebuild never wipes them. The **Composition** join picker offers only datasets visible in your *active* domain (My / Domain / Company of the operating domain, with the Marketplace as the only cross-domain surface); its **"Keep columns" starts with every column kept** – remove the ones you don't want, or *Remove all* and hand-pick. AI is built into each section as big ✨ **actions** – "✨ Draft documentation", the structured "✨ Clean it up" that fills the guided cleaning controls for your review (the AI never builds on its own), "✨ Explain this error" beside a real ingest error, "✨ Propose a clean/join", "✨ Suggest quality rules". The **Checks** section is a real data-quality gate: author dropdown-driven rule checks (`not_null` , `not_blank` , `unique` , `accepted_values` , `range`) that compile to SQL and run *for real* against the built table, yielding a **quality scorecard** – a 0–100 pass-rate and a passing/failing badge (honestly `unknown` , never a fake pass, when nothing ran) – let the OS **suggest rules from the column**

profile, and watch three heuristic **monitors** — freshness, row-volume and schema stability — that learn each dataset's normal band from its own run history. Opening a tile lands on the full-page **View: Talk to Data** (governed NL→SQL, one validated read-only `SELECT`, executed under your row filters) → a **preview** with a raw • cleaned • business layer toggle (business by default) → **statistics** → the **quality scorecard** → **configuration**. Until it is shared the header carries a "**not certified**" chip, and sharing (Promote • Certify) lives in that header, not a stage. A **Developer** view exposes the raw dbt SQL, the physical table and the **lineage** graph behind the guided flow. A table imported from an external warehouse doesn't go stale either: a "**Keep this in sync**" panel schedules regular refreshes — **Full refresh**, **Add new rows** (incremental append over a cursor column, with a late-data lookback window), or **Update by key** (a merge) — Hourly / Daily / Weekly or your own cron. Every run executes **as the dataset's owner** through the same governed Trino path as the original import, with the cursor predicate pushed down to the source — no data ever flows through the app — and each slice lands in Iceberg carrying `_loaded_at` / `_batch_id` lineage columns. The dataset shows its **sync history and watermark**; ten consecutive failures **auto-pause** the schedule (with a one-click *Reset & full re-sync* recovery), and freshness in **Monitoring** reflects the last sync. Scheduled sync is built for *incremental slices*, not bulk backfills: the current per-run limits (statement timeout, single-node Trino sizing, object-storage capacity), the honest estimates behind them, and the concrete scale-up playbook — exact Helm values, backfill windowing, and when to switch to staged-file loads — are documented in `docs/data-sync-scaling.md`.

- **Connections — governed bridges to outside systems.** A Connection is `credentials + endpoint + a set of governed tools`, never a raw pipe — used to bring data in and to expose external APIs/MCPs as tools. You grant **use**, never the token; **reads are automatic, writes are approval-gated** (destructive ops blocked), and secrets are write-only. The Supported Connectors gallery is **grouped by vendor stack** (Microsoft • Google • AWS • Databricks • Snowflake • Salesforce • Kajabi • Atlassian • Open source • Other) and searchable. When nothing in the gallery fits, a **Custom Connector** lets you add your own **REST/GraphQL API** or **MCP server** in one governed action: you name it, give the base URL and a write-only credential, and the OS **atomically files the egress-allowlist request** for that host — reads auto-allow, writes stay off until you enable them per-tool, and the far-side host cannot be reached until an Administrator approves the egress. The real catalogue now spans: **operational databases** (PostgreSQL • MySQL • SQL Server • MongoDB, federated through central Trino); **code & DevOps** (GitHub); **docs & knowledge** (Notion, Atlassian); a Supabase connector; **messaging & calendar** (Slack • Gmail • Google Calendar • Outlook • Teams — sending a message or email is always approval-gated, never automatic); **cloud governance / ML** (Microsoft Entra • Purview • Azure AI Foundry • AWS SageMaker, read-only); plus the established data-ingest (Google Drive / OneDrive), orchestration (Airflow) and catalog (**OpenMetadata**, read/discover of a customer's existing catalog, DLS-clamped to the tables the caller may already see — and now **active-only**: when you archive an OS dataset the integration best-effort **soft-deletes** its OpenMetadata entity, and unarchiving restores it, so the catalog reflects what's live rather than accumulating ghosts) connectors. Setting up the OAuth app / tokens on the far side is the **operator's step** (each connector ships an install guide). For teams already running a lakehouse elsewhere, an admin-enabled **external-warehouse** connector federates it through central Trino as a governed catalog — AWS

Glue/Athena, Snowflake, BigQuery, Databricks/Delta, and (experimental) Microsoft Fabric/OneLake – so you can query it in place under the same OPA path, or import a core table as a governed data product into the sovereign lakehouse – and keep the copy fresh with a scheduled sync (see *Data* above), so an import is a living dataset, not a one-time snapshot. Operational sources are first-class here too: PostgreSQL / MySQL / SQL Server sync on a timestamp or id cursor, **Kafka** topics land append-only on a per-partition offset cursor (deduplicate downstream), **Salesforce** objects sync incrementally by `SystemModstamp` over the REST API, and **Kajabi** resources sync over its public API with honest per-resource cursors (purchases incrementally by `updated_at`; contacts/customers/orders by `created_at` – new records only, edits need a full refresh; resources without a documented cursor are full-refresh only) – schedules run from every 15 minutes (append recommended at high frequency; frequent merges accumulate delete files) up to weekly. And for BI on your own desktop, a **one-click Power BI** button downloads a `.pbids` file that drops Power BI Desktop straight into the pre-filled PostgreSQL connector for the **Cube SQL API** – connecting as the `bi_<domain>` principal in **DirectQuery** mode, so per-domain row security re-runs on every query and no password is ever written into the file.

- **Metrics – one number, everywhere.** The KPI semantic layer, on the same one-artifact model: **New** picks a type, **Edit** defines it, and a full-page **View** reads it back. Define "Revenue" once – a **virtual declaration** the OS compiles into one governed Trino `SELECT` over the dataset's business (Gold) layer, run **as the viewer** – and it resolves to the *same* number in the explorer, in dashboards, and in an agent's `metrics` tool, each under the viewer's own row- and column-level security. **New** asks **Simple or Complex** on a two-card chooser first (the formula left the aggregation dropdown, where it was buried and undiscoverable). Tiles group into **Simple Metrics** (a single aggregation) and **Complex Metrics** – a composite built as arithmetic over this dataset's existing metrics with `[metric]` references and **null-safe division** (e.g. `([revenue] - [cost]) / [orders]`). A **materialized business layer is enough**: you can define, preview, explore *and* build a complex metric on a *personal* dataset with built Gold – no promotion needed; promotion is only to *share* it (and register its cube for dashboards). The metric View shows where it is already charted – an **On dashboards** section with **Add to a dashboard**. The builder's column palette shows the **actual business-layer columns**, including joined datasets' columns for a curated dataset; each definition emits a portable **dbt-MetricFlow-style semantic declaration** (`semantic/<slug>.yaml`) into the dataset's artifacts; and if the lakehouse is unreachable the metric honestly reads **unavailable** – never a fabricated number.

One folder UX on every context tab. The Context tabs – Data, Metrics, Files, Knowledge and Connections – all share the *same* folder experience (one core primitive, `lib/core/folders.ts`, with each tab registering a thin adapter – no per-tab divergence). Each tab shows a scope segment (My / Domain) and a single **folder rail tied to the active scope** – you only ever see the root that matches. You **create, rename and move** folders and items through a **folder-tree picker** (browse-and-click, with inline New-folder – never a text field); moving a folder carries its contents. Lifecycle is the shared one too: **Archive** a folder and it cascades to the items inside (with a warning – move items out first to keep them active); **Restore** or **Delete** (physical delete is archived-only, per-item permission-checked).

Talk to any Context tab. Every tab above carries a read-only "Talk to X" copilot. It builds a security-scoped overview of what *you* can see on that tab, runs the tab's own governed retrieval **as you** (Data → NL→SQL, Knowledge → knn retrieval, Files → file search; Metrics and Connections grounded on their catalog), and packs it within the model window via the **Context Assembler**. Answers arrive with the model's **reasoning shown separately** (a collapsible "thinking" panel), real citations, and a "what ran" disclosure – and it degrades honestly rather than inventing an answer when retrieval comes back empty.

4.4 Build

- **Agents – compose, govern, run.** A domain's **agent systems** (instructions + tools + memory) move through **five phases – Define • Design • Build • Run • Evaluate**. In **Define**, "**What your team can use**" is the interactive grants surface, and a grant here is a **default-on capability: every agent in the system inherits the full set of the system's Define grants by default** – you *narrow* an agent to give it less, never scramble to add. Per item you choose **read-only • read + propose • read + write** (a clear labelled selector), capped by the system's overall access setting; a **read + write** grant provisions that agent's matching **write tools** (e.g. `upload_file`, `create_dataset`), and a hard invariant guarantees an agent can never exceed the team's grants. Grants split into two groups – **Plan Items** (Strategy • Big Bets • Operating Model • Workflows) and **Context** (Knowledge • Files • Data • Connections • Metrics) – and **all four Plan Items are grantable**: granting a pillar or bet provisions its governed read tools (`get_pillar` / `get_big_bet`), DLS-scoped to what the caller may view. Context items grant via a **folder-tree with tri-state checkboxes** – tick a folder to grant everything in it (and future contents, resolved at run time, budget-capped, every resolved item still per-item DLS/OPA-checked so a folder grant is provably a *subset*), or tick individual items. Each **data grant** can target the **medallion layer** the team reads – Bronze, Silver, or Gold – and the picker only offers layers that are actually built, defaulting to the highest (Gold, the curated default). **Design** composes the agents three equivalent ways – a React-Flow graph builder, Monaco YAML editing, or a chat assistant. **Build** (*Build = execute + verify*) provisions and checks the team behind a live progress stepper; **Run** executes it as you and offers a "**Download PDF Results Report**"; **Evaluate** attributes context per agent and offers a "**Download PDF Evaluation Report**" – both fully brand-styled (gold-lotus cover, embedded datamasterclass fonts). Every call routes through **LiteLLM → OPA → Langfuse**.
- **Software – a governed frontend over the OS API.** An app here isn't a black box you bolt on – it's a first-class client of the OS itself. It moves through the shared five-stage builder – **Define** (state the purpose and grant the app its context) • **Design** (epics + user stories) • **Build** (the Simple view puts the app's *structure* first: the **Epics & stories tree** from Design, each story with an honest status chip – *to do • building • done • blocked* – and a "*N of M stories built*" count; **click a story to make it the build target** for the run, click it again to build the whole app. Beside it, a **live streaming build**: the AI streams its plan, then one honest, human-readable line per action – "*Committed 3 files*", "*Provisioning preview...*" – with errors shown as warnings with the real reason and retries visible in the feed, plus a real before/after **file diff** of what was committed per run; the raw code panel lives one click away in the **Developer view**) • **Test** (a "*Verify & Improve*" pass – the reasoning model checks each built story against its spec across five

dimensions — Functionality • User Experience • Code Structure • Security • Documentation — and turns any shortfall into a tracked refinement with a visible **Proposed → Designed → Built** state, beside a live in-cluster preview pod) • **Publish** (the Builder-reviewed deploy, promote / certify, the live tool surface, and lifecycle). A persistent **Build ▶ Preview ▶ Deploy status rail** keeps the honest, single-glance state in view throughout. At **Define** you pick one of **four scaffold templates** — **Application** (the default, full OS UX), **Website**, **APIs only** (no user interface), or **Empty app** — and each pre-shapes the epic structure the Build stage works through. The default is the **Sovereign standard app template** — a Vite + React + TypeScript SPA that already *is* an OS app before the first story is built. It boots *in the OS design*: it vendors `@sovereign-os/ui` (the gold-on-black AppShell and `.sb-*` primitives, no build step, no registry) and calls back into the platform through the **OS-client SDK** (`@sovereign-os/app-sdk` — `createOsClient().whoami()`, `.datasets.list()`, `.metrics`, `.knowledge`), so a brand-new app already renders *real governed data* under the signed-in user's own row security. Sign-in is **delegated to the OS session** — no local accounts or passwords, ever. The app shows its **owning-domain badge** and ships **My / Domain scope helpers** that filter every record by domain + user; an **Admin section** (OS `domain_admin` / Administrator only) carries a **read-only directory** of the OS users who can reach the app — managing users stays in the OS; and a top-bar **MCP button** links to the app's own MCP connection in Connections. The scaffolded **README is the build contract** — it documents how each story adds a page + section, and the Build assistant reads the same text as context. Each built story page is **auto-wired into the app's navigation**: the OS deterministically regenerates the section registry from the committed story pages on every commit, so a written story can never be left invisible ("builds fine but no feature shows"). Code commits to an in-cluster **Forgejo** repo (no GitHub, no tokens, your code never leaves). Commits are **sha-aware** — every write fetches the live blob sha first, so concurrent edits can never silently no-op or overwrite — and the pipeline status is **earned, not claimed**: the CI badge reflects the actual outcome of the latest Actions run on your commit (a missing repo secret even self-heals on the next push), and a build that *fails* is loud — the Test and Publish stages say plainly that the app is serving an *earlier* release with your recent changes undeployed, so a broken build never masquerades as "complete." An app can still **declare its surface** — `surface: ui | api | both` in `app.yaml`, which wins over auto-detection so a Streamlit/Gradio/Flask UI is never mislabelled "API." *Request deploy* assembles a review card — a security scan of the **live repo tree**, resource envelope, diff — that a human Builder decides in **Policies & Approvals**; on approve the in-cluster runner provisions a real Deployment + Service + Ingress with a live per-app URL (a `vite-os` app publishes as **static** files served by nginx). Apps carry the same lifecycle as every other tab — **Archive → Restore / Delete**.

- **Science — classic ML** (*opt-in, Layer 4*). Take traditional ML (regression, forecasting, clustering — *not* LLMs) from a governed data product to a deployed model-as-service, exposed as both a REST `predict` API and a `predict` MCP tool. Off by default; GPU is cost-gated.
- **Dashboards — governed BI, rendered natively**. Dashboards are built and rendered *in the OS* — **Apache ECharts on the governed Cube semantic layer** — and panels resolve the same metric declarations the Metrics tab serves, so BI and agents can never disagree. (Dashboards still query through Cube; the metric read path itself is now direct governed Trino SQL — a dual-run, with dashboard migration next. See *The lakehouse & semantic layer*.) They read the same one-

artifact model as the Context tabs — **New** → **Edit**, tile → **View**,  **Edit**, no stepper. In **Edit** you name it, bind **one governed Cube view** via metric chips, then design panels — metrics, dimensions, time grain, filters, per-panel width ($\frac{1}{3} \cdot \frac{1}{2} \cdot \text{full}$), and viz types led by **pie • bar • table** (then big number, line, area) — each with a live preview before **Save dashboard**. The full-page **View** renders every panel by querying Cube **as the viewer** (per-user row-level security, with a live/offline badge; switch *View as* and every panel re-queries as that viewer). The viewer surface adds **cross-filter chips** (click a bar or slice to filter the whole dashboard through a governed **WHERE**), a **drill-down drawer**, a **time-grain switcher**, persisted default filters, and **↓ CSV** on table panels. Any panel also **expands full-screen** (its title or ) — a graph shown large with its underlying rows as a table and **↓ CSV** beneath, a table panel simply rendered large. Sharing lives in the **header** — promote / certify, schedule reports, and **Connect tools**, not a stage. Connect tools are the Tier-2 BI bridge over the **Cube SQL API** as a domain-scoped read-only principal: **one-click Power BI** (a pre-filled **.pbids** file), **Tableau** connection fields, and — when the operator has configured it — an **"Open in Superset →"** link to Superset's own console in a new tab, never embedded.

4.5 Monitor & Admin

- **Governance** (*Builder+*) — the control plane: one Approvals inbox for every side-effectful action, the consolidated policy view, the hash-chained audit, cost caps, and Users & access.
- **Monitoring** (*Builder+*) — artifact observability, scoped to your identity and strictly read-only. Two tile boards, each My / Domain / Company: **Agent Monitoring** — every agent system with its real last-7-day telemetry (runs, last run, warnings/errors, and **Tokens truly measured**, captured from the model gateway per run; **Cost** appears only when model pricing is configured via **MODEL_PRICES_JSON**, otherwise an honest "-", never a fake 0) — and **Data Monitoring** — every dataset with freshness, pipeline health and its DQ status (a red *"N DQ rules violated"* is the cue). Open any tile for the full diagnosis view — it takes over the main window like every other tab's detail: run history with per-node drill-down, the system profile (agents, models, grants), governed tool-call traces, cost/token trends, and for datasets a **Data-Quality dashboard**. It also rolls up **data quality** across your datasets: a risk-ranked board (riskiest first) built from the same quality-run history, a domain health average, and an honest count of datasets that have **never been checked** — surfaced as a gap, not painted green.
- **Console** (*Builder+*) — the governed **Query** surface: Lakehouse SQL runs through Trino under the caller's own OPA row/document-level security, so a builder can explore data safely. The **raw Shell** and the unscoped Cube query mode stay **admin-only** (in the UI *and* the API). For developers who'd rather stay in their own terminal, the same governed door is a CLI: **sos** (**cli/sos/**, Phase 0) is a thin Go client that signs in with OAuth 2.1 PKCE and then runs **whoami**, **datasets list** / **get**, and **query "<NL or SQL>"** (or **--metric**) — **every call over the OS MCP front door, as you**, with role, domains, OPA and row/document security re-resolved live on the server. It holds only a short-lived token in your OS keychain; there is no privileged side-channel.
- **Components** (*Admin*) — the one operator surface: every platform service with live health and version, and one-click same-origin consoles (Superset, Forgejo, Dagster, ...) via SSO.

- **Admin** (*Builder+, filtered*) – the tenant control room (domains, users, models, egress, cost, backups). A builder sees the tab too, filtered to a single **My Settings** self-service tile; every tenant-admin tile stays admin-only and hidden, and deeper admin sub-pages redirect non-admins back (fail-closed, default-deny). **About** carries the license inventory.

5 The golden paths — walked end to end

Abstract governance is easy to nod at and hard to feel. So here is the OS at work on a single, concrete story that runs through the whole guide: **Northpeak Commerce**, a fictional mid-sized European omnichannel retailer, whose team wants to optimize marketing-campaign budget with agents. (It's the exact case study seeded into the live teaching cohort — see *The live teaching cohort* — so every step below is a real, governed flow, not a mock-up.)

5.1 Golden path 1 — Data: from a CSV to a queryable metric

Meet **Mara**, a Creator in the `sales` domain. She has a `campaign_master.csv`.

1. **New → ingest.** In **Data**, Mara clicks **New →  Ingest new data**, names the dataset, and uploads the CSV in the **Ingestion** section. Her bytes land as a real Iceberg table in her *own* per-user schema (`iceberg.personal_mara.bronze_campaign_master`) — registered only when apply **and** a governed verify both pass. No fake green ✓.
2. **Clean it (Transformation).** In the **Transformation** section she applies guided ops (cast types, drop dupes, set the key) — or lets "✨ Clean it up" propose them into the same controls for her review — then **Save Transformations**. The OS compiles **one** allowlisted CTAS into her schema and runs it as her — OPA masks every read. Because this is a single ingested source, the business layer **materializes automatically via pass-through**: there is no separate "build Gold" step, and her dataset is already **metric-ready**, even before any promotion.
3. **Curate — join margin & CAC.** To bring in margin and CAC, Mara clicks **New →  Create a curated dataset** and, in **Composition**, picks the campaign dataset as the explicit **base**, joins margin and CAC on a reconciled key (the join picker offers only her active domain's datasets), keeps the columns she wants and adds a derived field — then **Save Composition**. The composed business table materializes; measures come later in Metrics, so a rebuild never wipes them. (A single ingested dataset never joins — the curated path is where joins live.)
4. **Checks — quality & lineage.** In the **Checks** section Mara authors a few rules (`not_null` on the key, `unique` on the campaign id, an `accepted_values` list for `channel`, a `range` on `spend`) — or lets the OS **suggest them from the column profile** — and **Save Data Quality Checks** runs them for real against the built table. She gets a **quality scorecard** and a passing badge, plus freshness/volume/schema **monitors** that will flag the table if its next load arrives late, comes in thin, or changes shape. The **View** (and the Developer lineage graph) shows exactly where every number came from.
5. **Document & promote.** Documentation (the **Documentation** section) is the gate: Mara adds a description and column docs, then promotes from the dataset **header** — until then it carries a **"not certified"** chip. She's a Creator — she *cannot* approve her own work, so this files a request.
6. **A Builder approves — and the publish runs.** **Ben**, a Builder in `sales`, approves. The approval independently verifies the physical business table materialized in the domain schema (`iceberg.sales.gold_campaign`), then flips the tier and writes the audit.

7. **One number, everywhere.** In **Metrics**, `revenue`, `aov`, `conversion_rate` and `churn_rate` now resolve on the business layer, sliceable by `region`, `product` and `date` – no SQL. Each metric is a declaration compiled into one governed Trino `SELECT` run **as the viewer**, so anyone who asks "what's revenue?" – a dashboard, an agent, the explorer – gets the same answer, under their own row filters. (Mara didn't even have to wait for step 6: a materialized business layer of *any* tier is metric-ready, so she could have defined and previewed these on her personal dataset – promotion is what shares them and registers the dashboard cube. See *The lakehouse & semantic layer*.)
8. **Talk to it.** In the dataset's **View**, Mara opens **Talk to Data** and asks a plain-English question. The model is shown only datasets she can see, generates one validated read-only `SELECT`, executes it through governed Trino under her masks, and answers grounded only in the returned rows.

(This entire path is also available over MCP – `create_dataset` • `ingest_dataset` • `transform_silver` • `build_gold_join` • `document_dataset` • `request_promotion`, then a Builder's `approve_promotion` runs the physical publish. Same governed functions, no back door.)

5.2 Golden path 2 — Knowledge: capturing how the work is done

Northpeak's campaign playbook lives in people's heads. Let's make it retrievable.

1. **Author a workflow.** In **Knowledge**, Ben authors a "*Campaign budget decision*" workflow: ordered **steps** (each owned by a Human / Software / Agent actor, with inputs/outputs), **business rules**, and **expert knowledge** – the gotchas and the "why behind the why," which get indexed as first-class retrieval units.
2. **Mark a hard rule.** "*CAC above target for 14 days ⇒ never INCREASE budget*" is marked **hard**, so it compiles into an OPA guardrail an agent must respect.
3. **Index & verify.** `index_knowledge` chunks and embeds the workflow into OpenSearch; a quick `search_knowledge` confirms it surfaces. Indexing is *not* automatic – this step is what makes it findable.
4. **Publish.** A Builder publishes it to **Domain** scope, so every domain agent can ground on it. Expert-knowledge notes carry provenance, and agents must cite the source.

5.3 Golden path 3 — Agents: a governed team with real hands

Now the payoff. Mara builds an agent team that reads the campaign data, respects the playbook, and recommends a budget next-best-action.

1. **Compose.** In **Agents**, Mara drags an *analysis* agent and a *recommendation* agent onto the React-Flow canvas (or edits `system.yaml` directly – same versioned file). Each agent's `AGENT.md` grounds it in the published campaign knowledge.
2. **Grant resources + tools.** In "What your team can use" she grants the system the Domain-scope campaign datasets, the knowledge workflow, and the `query_data` / `search_knowledge` tools, each at read-only. A validation gate must pass; a sub-agent's grants are always a strict subset of the system's.

3. **Pick models.** The single **Auto / Reasoning / Execution** toggle shows the real gateway model names (`sovereign-reasoning` , `sovereign-default`) with an internal/external badge.
4. **Build = execute + verify.** *Build* runs the compiled system and checks it – every call routed **LiteLLM → OPA → Langfuse**.
5. **Run – governed all the way down.** *Run* executes the team **as Mara**. Every tool call the team makes dispatches through the same governed door as her own MCP calls: grant-scoped, OPA-pre-gated, role-floored. The team returns *INCREASE / CUT / HOLD budget for X days + reasoning*. A write pauses for approval and enqueues in **Governance** – the agent is *propose-don't-commit* by default.
6. **Evaluate – see what each agent actually used.** The **Evaluate** view attributes context *per agent*: exactly which datasets, docs, files, metrics and connections each agent read, how (tool + read/retrieved/written + a short args hint), each a **clickable deep link** that opens the real artifact (switching scope so it's visible). A granted-vs-used strip flags dead grants. So Mara can prove the team grounded on the campaign knowledge, not on thin air.
7. **Promote.** Once it's good, Mara files a promotion; a Builder shares it (to Domain) so the whole domain can *run* it (but not edit it).

Two runtimes, one governed plane. A system picks **LangGraph** (the default – structured, replayable, human-in-the-loop) or the autonomous **Hermes** runtime for long-running work that compounds (persistent memory + self-improving skills). Both share one governed plane: Hermes reaches models **only** through LiteLLM and tools **only** through the same Platform MCP, so OPA still gates every call, Langfuse traces it, and code runs in a kernel-isolated sandbox (Kata microVM or gVisor – never host-local). Hermes ships **off by default**.

5.4 Golden path 4 — Big Bets & Strategy: tying it to value

Finally, the work connects to the plan. In **Strategy**, an Administrator defines a pillar – *"Marketing efficiency"* – with a value metric (say, EBIT, or a custom "CAC reduction %"). In **Big Bets**, Ben creates a *"Campaign budget optimization"* bet under that pillar, sets a target and a go-live date, and attaches the real artifacts built above – the Gold dataset, the metric, the agent team, the app. Status derives **live** from each artifact's real lifecycle, so the roadmap flags itself on-track or at-risk without anyone updating a spreadsheet. The pillar's value can be tracked as a governed metric or entered monthly – either way it feeds the pillar's history chart.

6 The governance model — the honest details

Governance here is not a policy PDF; it's executable, and it's the same for a click and for an agent. Four guarantees hold throughout.

6.1 Run-as-user, always

The per-user token (UI session or MCP token) carries your identity. The **role floor is re-checked from the live session on every call** — never trusted from the request body. An agent runs as its owner; Ask the OS runs as you. Nobody and nothing gets a privileged path.

6.2 OPA tool gates

A principal may invoke a tool only if **granted**; unknown principals and ungranted tools are denied. Internet access is an explicit grant — `web_fetch` is ungranted by default, and the web is returned as **sanitized data, never instructions**. Policy checks **fail closed**: if OPA is unreachable, the gate *denies* (with an explicit `opa-unreachable` marker) rather than waving the call through.

6.3 DLS — row & column security, independent of tier

Document- and row-level security filter what you see **at query time, regardless of tier**. Promoting an artifact to a wider scope **never** widens row access — two viewers of the same Domain-scope dataset see different rows. Every data-proxy route requires a session and scopes results to the caller's domains.

6.4 Two policy layers, one inbox

- **Tenant guardrails** an Administrator sets and domains cannot override: default-deny egress, no plaintext secrets to agents, no cross-domain data without a grant, a model allowlist.
- **Domain policy** Builders set within them.

High-stakes actions don't fail silently — they queue as a **card** in the **Governance** inbox, where *approving is the action*: on approve the platform runs the effect (an Argo deploy, a policy grant, an egress allowlist entry, a promote, a queued run) and writes the hash-chained audit. Three planes stay deliberately separate and cross-link rather than duplicate: **Admin configures** the tenant, **Governance decides and records**, and **Monitoring observes the artifacts**.

6.5 Archive, delete, and version history

Every artifact carries the same lifecycle controls, and only its owner (or an in-domain Admin) may use them. **Archive** is a reversible soft-hide (a running agent is also stopped); **Delete** is available only on already-archived items, behind an explicit confirm; **Version history** snapshots the prior state on every meaningful edit, and *restore* itself snapshots the current state first – so you can always undo a restore. One reusable helper (`lib/core/versioning.ts`) gives every store identical behaviour, and history is mirrored to OpenSearch so it survives redeployments.

7 The architecture

The platform assembles ~two dozen open-source components you can reason about – and enable – in layers. On top sits the **OS UI**, the single front door; beside it, the **MCP servers**, the governed front door for AI.

flowchart TB

```
subgraph front["Front doors – one governed path"]
    UI["OS UI (Next.js)<br>server-side routes"]
    MCP["MCP servers<br>/api/mcp + /api/mcp/&lt;tab>"]
    EXT["Claude · ChatGPT · any<br>Streamable-HTTP MCP client"]
    EXT --> MCP
end
```

```
UI --> SPINE
MCP --> SPINE
```

```
subgraph spine["The governed spine – lib/infra/governed.ts"]
    SPINE["authorize → act → trace"]
    OPA["OPA<br>policy-as-code"]
    LF["Langfuse<br>trace + cost"]
    SPINE --> OPA
    SPINE --> LF
end
```

```
SPINE --> L1 & L2 & L3 & CTX
```

```
subgraph L1["L1 – Agent core"]
    LG["LangGraph agents"]
    LL["LiteLLM gateway<br>(budget cap, allowlist)"]
    OS["OpenSearch<br>(hybrid retrieval)"]
    LG --> LL
    LG --> OS
end
```

```
subgraph L2["L2 – Foundations"]
    CUBE["Cube (dashboards)"]
    DBT["dbt · Dagster"]
    OM["OpenMetadata"]
    DOC["Docling · Haystack"]
end
```

```
subgraph L3["L3 – Self-service"]
    TRINO["central Trino"]
    ICE["Iceberg / Polaris / MinIO"]
    SUP["Superset (optional BI console)"]
    FORGE["Forgejo + Argo CD"]
    TRINO --> ICE
end
```

```
CTX["Context Assembler<br>budget-aware prompt packing"]
LL --> STK["STACKIT AI Model Serving<br>(EU-sovereign models)"]
```

7.1 The layers

- **Layer 1 – Agent core.** The runtime: **LangGraph** agents calling **LiteLLM** (the one model + tool gateway, with per-key access control and cost caps), every action traced in **Langfuse**, retrieving over **OpenSearch** (hybrid vector + lexical – no separate vector DB).
- **Layer 2 – Foundations.** Turning raw data and knowledge into governed products: **OPA** (policy at the tool boundary), **Docling** (parsing), **Haystack** (RAG), **Dagster** (orchestration), **dbt** (transforms), **Cube** (the dashboard query layer – metrics themselves compile to governed Trino SQL, see below), **OpenMetadata** (catalog + lineage).
- **Layer 3 – Self-service.** Query, visualize, ship: the **Iceberg** lakehouse (**Polaris** catalog, **MinIO** object storage) with **central Trino** as the *one* governed query engine, **dashboards rendered natively in the OS** (Apache ECharts on the Cube semantic layer; **Superset** remains an optional stand-alone console), and in-cluster **Forgejo + Argo CD** for software delivery (git → CI → GitOps).
- **Layer 4 – Science / ML.** Classic ML – **JupyterHub**, **MLflow**, **Featureform**, **KServe** – *opt-in and off by default* (heavier, GPU-oriented).
- **Security baseline** spans every layer: default-deny egress through a single proxy chokepoint, a governed `web_fetch`, OPA tool authorization, externalized secrets, hardened pods.

7.2 The lakehouse & semantic layer

An upload becomes a real **Iceberg** table in your per-user schema; Silver and Gold builds run one compiled CTAS each; everything is queried through **central Trino** under your identity, so there is exactly one governance boundary for data. **Polaris** holds the catalog metadata in a durable relational-JDBC metastore (so the warehouse registration survives restarts), and **MinIO** keeps the data files on a PVC. Above the lakehouse sits the semantic layer – and here the OS runs an honest **dual-run**, mid-migration. **Metrics are served by direct governed Trino SQL – Cube is off the metric read path.** A metric is a *virtual declaration* the OS compiles into one governed `SELECT` over the physical Gold mart and runs **as the viewer**, so Trino/OPA row- and column-level security applies and every result is honestly labelled *live (sql)*. Because the read path is plain governed SQL, a **built Gold of any tier is metric-ready**: a personal dataset's metric reads the owner's private lane (`iceberg.personal_<owner>.gold_<slug>`) as the owner; a governed dataset reads the domain mart – promotion is needed only to *share* a metric and to register its cube for dashboards (the gate is split so no cube is ever registered on a personal Gold). Each `define_metric` also emits a portable **dbt-MetricFlow-style semantic declaration** (`semantic/<slug>.yaml`: the semantic model with the Gold-mart ref, primary-key entity, join-aware dimensions with time grains, and measures + metrics) into the dataset's artifacts – the tool-agnostic contract the compiler serves as Trino SQL. And the honesty gate holds end-to-end: if the query backend is unreachable on a real deployment, a metric returns an honest *unavailable* – never a fabricated number. **Cube stays running for dashboards only** (Phase 2 migrates those): a promoted Gold dataset still auto-registers as a queryable Cube model, and Cube picks up new and changed models **deterministically** – its `schemaVersion()` hashes every model file's name and bytes, so any add/edit/remove triggers a lazy, per-context recompile on the next query, without a restart. One honest exception: rolling-window and running-total measures have no SQL form yet, so they serve via Cube post-Publish until Phase 2.

7.3 Models & the gateway

Every model call goes through **LiteLLM** – the one gateway that enforces the allowlist, per-key spend caps, tracing and graceful back-pressure. Inference runs on **STACKIT AI Model Serving**, an EU-sovereign, pay-per-token, three-tier set that an Administrator configures in **Admin → Models & Providers** (a single live-sourced store; the three below are the helm defaults – the OS is admin-configurable, not hardcoded to any provider). Each model also carries **per-token prices (EUR per 1M input / output tokens)**, editable in the same Models & Providers screen; these drive the cost figures in Monitoring – a model with no price set shows "–" honestly rather than a fake €0:

ROLE	HELM-DEFAULT MODEL	LITELLM NAME
Reasoning / planning (the PLAN phase)	Qwen3-VL-235B-A22B-Instruct-FP8	sovereign-reasoning
Standard / worker (tool-calling, coding, chat)	gpt-oss-20b	sovereign-default
Embeddings (4096-dim)	Qwen3-VL-Embedding-8B	sovereign-embed

STACKIT usage draws on one shared **€250/week** budget; once exhausted the gateway returns a graceful HTTP 429 rather than failing hard, and resets weekly.

7.4 The Context Assembler

Agents that read real data hit real context limits. The **Context Assembler** (`lib/infra/context/`) is a budget-aware prompt builder: a per-model window registry (with reserved output, admin/env-overridable), tool-result **compaction** (row-sets → header + sample + "...N more"; long text → head/tail), and a greedy pinned-first pack that **guarantees the prompt never exceeds the model window**. It's wired into the single-agent harness, the multi-node graph handoff (each node hands on an assembled summary, not the full transcript), *and* Ask the OS – which is what lets an agent discover and query real tables without blowing a 200K window.

7.5 The MCP front door

The platform exposes itself as **governed MCP servers**, live end-to-end at

`https://agentic.datamasterclass.com/api/mcp` – one cross-tab server plus per-tab servers at `/api/mcp/<tab>` (`software` , `data` , `knowledge` , `agents` , `files` , `metrics` , `dashboards` , `bigbets` , `science`), each shipping a token-minimal `CONTEXT.md` . Around **55 governed tools** ship – reads, writes, and read-back parity – so an external client can build the entire Data → Metrics → Agents flow above. Every tool delegates to the **same library function the UI calls**; promotion-class actions stay Builder/Admin-gated; and every failure returns a typed, model-readable `{ code, reason, hint }` so a client can self-correct.

7.6 Durability

Every user-facing in-process store mirrors to **OpenSearch** through one shared core

(`lib/infra/os-mirror.ts`): write-through on change plus hydration on boot, so artifacts survive redeploys and node-rolls. On STACKIT a three-tier backup system (nightly Postgres dumps, nightly off-cluster Velero volume backups, and a pre-upgrade backup gate) protects the stores themselves – practiced with a restore-drill runbook, with honest documentation of what is *not* protected.

8 Quickstart — run it

8.1 Locally, in one command

```
# prereqs: docker (running), kind, helm, kubectl · ~14 GB RAM / 6 CPU free
./install.sh                # press Enter through every prompt
```

Pressing **Enter** through every prompt gives the **fully self-contained** install: every backend runs inside the chart and a tiny local model answers model calls — nothing external, no API key.

`install.sh` creates the `kind` cluster if needed, builds and loads the images, installs the chart, seeds the demos, and prints the front door and demo logins.

```
./install.sh --defaults    # non-interactive, all bundled (CI / quick)
./install.sh --uninstall   # remove the release (keeps the cluster)
```

8.2 Open the front door

```
kubectl -n agentic-os port-forward svc/os-ui 8080:3000 # → http://localhost:8080
```

Every surface calls the in-cluster backends through **server-side API routes**, so credentials never reach the browser. Locally there's no login; on a real deployment you sign in with your Ory identity. The stack's operational console is embedded at **Platform → Components** — there's no separate admin service to run.

8.3 Try the seeded demos

Four end-to-end demos ship seeded, so the system proves itself the moment it's up: **ask the RAG agent** (retrieve → generate → trace), **query the lakehouse** (the governed `query` tool over central Trino), **build a dashboard** (native ECharts panels on governed metrics), and **ship software** (push → Forgejo CI builds an image → Argo CD redeploys). Each has a one-card launcher on **Home**.

8.4 Deploy to your cloud (STACKIT)

The same chart runs the full Layer 1–4 stack on a sovereign STACKIT cluster; switching a backend to a managed service is only a values choice. Provision an SKE cluster + Object Storage + a load balancer + DNS, bootstrap the in-cluster prerequisites, point the OS at managed backends in

`values.stackit-managed.yaml`, then:

```
helm install agentic-os charts/sovereign-agentic-os -n agentic-os --create-namespace \
-f values.stackit-managed.yaml -f values.generated.yaml
```

Full, verified steps live in the **Reference** below and in [docs/stackit-deployment-guide.md](#). Rough cost: **€450–670/mo** for L1+L2 at typical sizing; scale the node pool to zero between sessions (storage + IP persist at ~€16–20/mo).

8.5 Connect from Claude or ChatGPT (MCP)

Point any Streamable-HTTP MCP client at <https://agentic.datamasterclass.com/api/mcp> (or open any tab's **"Connect your AI Tool via MCP"** button for a one-click import link). The server uses managed OAuth (client-id-metadata-document pattern): your client fetches the metadata, you approve once at the OS consent screen, and it receives a 180-day access token held in the client — never in the OS. From then on **every tool call runs as you** — role floor re-checked from the live session, OPA-authorized, DLS-scoped. First stop, always: `whoami` and `list_capabilities`.

9 How to contribute

The OS UI is where most contribution happens, and it's built to be joinable. One rule governs the whole layout: **everything is either a tab, infrastructure, or core**. Learn one tab and you can work on any tab, because every tab is shaped the same way.

9.1 The three layers

Dependency direction is strict and one-way: `<tab>` → `infra` → `core`.

- `lib/core/` – cross-cutting primitives (session, config, auth, scopes, lifecycle, versioning, the artifact model, nav). No tab logic, no external IO.
- `lib/infra/` – the governed spine + every external-service client. The *only* layer that talks to OPA, Trino, OpenSearch, LiteLLM, MinIO, Forgejo, k8s. `governed.ts` (authorize → queryRun → trace) is the spine every tab write goes through; `mcp/` is the MCP transport.
- `lib/<tab>/` – one module per OS tab, all with the same internal shape. A tab imports *down* into core + infra, and **never sideways** into another tab's internals – only through that tab's `index.ts`. (A tab reaching into another tab's internals is the one thing code review rejects.)

9.2 The tab-module contract

Every `lib/<tab>/` has the same files:

FILE	RESPONSIBILITY
<code>index.ts</code>	The tab's public API – the only thing other tabs / routes import.
<code>schema.ts</code>	The tab's types (artifact shape, tiers, visibility). Pure.
<code>store.ts</code>	The governed adapter – CRUD/list/promote/lifecycle, each through <code>infra/governed</code> .
<code><feature>.ts</code>	Pure, unit-tested domain logic. IO is injected so it stays testable.
<code>*.test.ts</code>	Co-located with the file it tests.
<code>README.md</code>	One screen: what the tab does, its golden path, its public API, its invariants.

Where to start. Read `os-ui/ARCHITECTURE.md` (the full contract) and `lib/connections/` (the reference tab-module: index / schema / store / README). Then pick a tab, copy the contract, and keep the invariant sacred: **all authz + trace live in `infra/governed` and each tab's `store.ts` – never scattered**. Consistency here *is* robustness: it's what makes the single governed path auditable.

See `CONTRIBUTING.md`, `GOVERNANCE.md`, and `CLA.md` at the repo root for the project process.

IO Reference

10.1 Deploying to STACKIT — the verified path

Locally everything is self-contained. On **STACKIT** (or any cloud) the platform runs the full Layer 1–4 stack from the **same chart**; switching a backend to a managed service is a values choice (`values.stackit-managed.yaml`), and heavier layers (Science, Terminal) ship pinned and provisioned but off by default.

1. **Prerequisites.** A STACKIT organization + project in **EU01 / Deutschland Süd**, and a service-account key with provisioning roles (SKE + Object Storage + DNS), saved as `stackit/sa-key.json` (gitignored). **This key gates any live deploy** – you can build and validate the entire chart on local `kind` with no key.
2. **Provision managed resources** (Terraform preferred): an **SKE cluster** (CNI = Cilium), a node pool sized for the full stack, **Object Storage** buckets + S3 credentials, a **load balancer + public IP**, a **DNS zone**, and **Secrets Manager / KMS**.
3. **Bootstrap the in-cluster platform** before the OS chart: ingress-nginx + cert-manager, the SKE storage class, Cilium default-deny egress, the External Secrets Operator, CloudNativePG, Velero, and Argo CD.
4. **Point the OS at managed backends** in `values.stackit-managed.yaml`: object storage and Postgres to STACKIT, the LLM to STACKIT AI Model Serving (`llm.mode: external`), Trino → the Polaris REST catalog, plus ingress hostnames, the egress allowlist and per-domain quotas. You can mix freely – managed Postgres but bundled OpenSearch, for example.
5. **Deploy and verify:**

```
helm install agentic-os charts/sovereign-agentic-os -n agentic-os --create-namespace \
  -f values.stackit-managed.yaml -f values.generated.yaml
```

Point DNS at the load balancer, confirm the consoles, confirm the default-deny egress baseline is active, then create your first domains.

Recommended: single node. The primary, verified STACKIT path is one node, single AZ, all backends self-contained (`docs/stackit-deployment-guide.md`). Managed-services mode and multi-node HA are currently known-blocked on SKE cross-node networking; single node sidesteps it.

10.2 First-run bootstrap

On a real deployment the platform starts **closed**. The secure first-run path:

1. **Claim the first administrator.** Identity is backed by **Ory**; the bootstrap creates exactly one tenant Administrator that **auto-verifies** — no email server required to start.
2. **(Optional) wire up email** — Microsoft Graph `sendMail` (recommended for M365) or an SMTP fallback; sender `support@datamasterclass.com`. With neither, the platform runs without email and later accounts are verified out of band.
3. **Create domains** (one per team/business area; toggle optional layers like Science).
4. **Invite real users** — by email (the email is the username), with a role per domain. The platform generates a one-time temporary password shared out of band; the invitee sets their own on first login.
5. **Set the guardrails** — model allowlist, egress allowlist, cost envelope — all compiling through the same OPA the tabs enforce.

Honest status on the live STACKIT tenant: outbound mail is currently **not delivering** (provider port-25 block; relay + sender-domain DNS pending), so on that deployment accounts are verified out of band. Graph and SMTP transports work wherever those services are reachable.

10.3 Sizing & capacity

Three different resources get confused under one word, "size":

RESOURCE	THIS DEPLOY	HOLDS	HOW IT SCALES
Node RAM	128 GB (<code>m3i.16</code>)	Running pods (Trino heap, OpenSearch; inference is on STACKIT, off-node)	With concurrency/workload , not data. Ran ~2–4% here.
Node disk	200 GB	Container images (all L1–4 images ~40–60 GB + churn headroom)	FIXED. Does not grow with your dataset.
Data storage	Object storage + PVCs	The Iceberg lakehouse on object storage (→ TBs) + PVCs for OpenSearch, Postgres, ClickHouse, MLflow	Independently , with the dataset.

The one gotcha: **don't confuse node RAM (128 GB) with the node disk** (the small volume that fills). Real data never touches the node disk — it lives on separate, independently-scalable storage.

10.4 Security model — four guarantees

- **Default-deny egress.** NetworkPolicies deny outbound except DNS, intra-namespace, and the API server; only the allowlist-only, logging **egress proxy** may reach the internet. (On `kind` the app-layer chain OPA → proxy → `web_fetch` provides the guarantee; on STACKIT, Cilium enforces it with FQDN-aware allowlists.)
- **OPA tool authorization, least privilege.** A principal may invoke a tool only if granted; agents use a scoped virtual key with a spend cap, never the master key.

- **The web is data, not instructions.** The only path out is the governed `web_fetch` – OPA-authorized, routed through the egress proxy, returned as sanitized data, never auto-written into the knowledge base.
- **No real secret in git.** On STACKIT every secret lives in Secrets Manager / KMS, synced by the External Secrets Operator; the chart references secrets by name only. The local dev passwords below exist only under `profile: local`.

10.5 Components at a glance

LAYER	COMPONENTS
L1 – Agent core	LiteLLM (gateway → STACKIT three-tier set) • OpenSearch (retrieval) • Langfuse (tracing) • query-tool (Trino MCP) • system agents (Domain RAG • ML pipeline • Hermes runtime)
L2 – Foundations	OPA • Docling • Haystack • Dagster • dbt • Cube • OpenMetadata
Infra	Postgres (CloudNativePG) • ClickHouse • Valkey • MinIO (PVC-backed) • Polaris (durable JDBC metastore)
L3 – Self-service	central Trino • Superset (optional console) • Forgejo (sovereign git) • Argo CD • CI runner • OpenSearch Dashboards • Terminal
L4 – Science	JupyterHub • MLflow • Featureform • KServe (opt-in)
Security & platform	egress-proxy • web_fetch • WireGuard tunnel (optional) • OS UI (embedded Components console • same-origin tool proxy + Level-1 SSO • MCP servers)

10.6 Demo logins (profile `local` — throwaway, never reused on STACKIT)

CONSOLE	PORT-FORWARD (<code>KUBECTL -N AGENTIC-OS ...</code>)	URL	LOGIN
OS UI	<code>port-forward svc/os-ui 8080:3000</code>	<code>http://localhost:8080</code>	—
Langfuse	<code>port-forward svc/agenti-c-os-langfuse-web 3000:3000</code>	<code>http://localhost:3000</code>	<code>admin@datamasterclass.com</code> / <code>langfuse-local-dev-admin</code>
Superset	<code>port-forward svc/agenti-c-os-superset 8088:8088</code>	<code>http://localhost:8088</code>	<code>admin</code> / <code>superset-admin-local-dev</code>
Forgejo	<code>port-forward svc/forgejo-http 3001:3000</code>	<code>http://localhost:3001</code>	<code>gitea_admin</code> / <code>forgejo-admin-local-dev</code>
Argo CD	<code>port-forward svc/argocd-server 8082:80</code>	<code>http://localhost:8082</code>	<code>admin</code> / <code>secret</code> <code>argocd-initial-admin-secret</code>
MinIO	<code>port-forward svc/minio 9001:9001</code>	<code>http://localhost:9001</code>	<code>agenti-c-os-local</code> / <code>agenti-c-os-local-secret</code>

(The full console table is in `docs/getting-started.md`.)

10.7 The live teaching cohort

The live STACKIT deployment doubles as the classroom for the **Agentic Leader Program**, and its setup is a worked example of the whole operating model. A cohort domain hosts the instructor as **Builder** plus the participants as **Creators** (each signs in with email as username), with a separate `test` domain for dry-runs. The **Northpeak Commerce campaign-optimization exercise** — the running example throughout this guide — is seeded at **Domain scope** through the platform's *own governed endpoints*: campaign datasets, knowledge documents, sample files, a ready-made Campaign Evaluation Agent, and a Campaign App. Because the materials are Domain-scoped, every participant can *use and run* them — but as Creators they cannot edit them or promote their own work without a Builder. The exercise teaches the promotion ladder by living inside it.

10.8 Troubleshooting

- **ImagePullBackOff** on **demo-app** **right after install** – expected; clears once the first CI run builds and bumps the image tag.
- **Out of memory / pods pending** – the local slice is RAM-bound; keep heavy components off or give the VM more RAM.
- **Agent answers look canned** – STACKIT AI Model Serving isn't wired (no **STACKIT_API_KEY**), so chat has no live model. Configure the key or point LiteLLM at any model – no agent change.
- **web_fetch** **returns 403 / 502** – 403 means OPA hasn't granted **web_fetch**; 502 means the domain isn't on the egress allowlist. Both are by design.
- **Do I have to use STACKIT?** No – any Kubernetes works; the chart is portable. STACKIT is the sovereign EU default. You can build and validate everything on **kind** with no cloud key.

10.9 Status — what's live, what's next

The governance spine – OPA, approvals, RLS, promote ladders, roles, audit, MCP (live end-to-end at **/api/mcp**), auth, Knowledge, and the physical Data pipeline (ingested or curated datasets: ingest → optional transform → automatically materialized business layer, or a curated composition of trusted datasets, with real data-quality checks + freshness/volume/schema monitors in the Checks section, then publish-on-approval → governed metrics → Talk to Data) – is **fully live**. The Context tabs and Dashboards share **one artifact model** (New → type → Edit sections with per-section Save → full-page View, honest state, an in-flow assistant, Simple/Developer views, lifecycle in the header); the systems tabs – Agents, Software, Science – keep a **staged builder** for their ordered deploy gates. Layers 1–3 are in place; **Science (Layer 4)** is an integrated model-as-a-service tab (Define → Train → Deploy → Predict → Monitor) wrapping a live KServe **predict** model, with the raw MLflow/Featureform/JupyterHub/KServe consoles as a Developer escape hatch. **Software** is now a *governed frontend over the OS API*: new apps scaffold from the **Sovereign standard app template** – a Vite/React SPA that boots in the OS design (**@sovereign-os/ui**), signs in through the OS session only (no local passwords), scopes records My / Domain, and calls back through the OS-client SDK (**@sovereign-os/app-sdk**) under the signed-in user's own security – with a **live streaming Build** (the plan first, then one honest line per action, warnings and retries visible, behind a persistent Build ▶ Preview ▶ Deploy status rail) driven from the **Epics & stories tree** (per-story status chips, one-click build targets, "*N of M stories built*") – every built story **auto-wired into the app's navigation** (the section registry regenerates from the committed pages, so a story can never build-but-not-show) and a **failed build surfaced honestly** in Test and Publish (never a fake "complete" over a stale release). It shows real per-run file diffs, live preview, and a Builder-reviewed deploy that scans the live repo tree; apps build a real image in-cluster (Forgejo CI) or publish static, and deploy to a live per-app URL. **Metrics** are served by **direct governed Trino SQL** compiled from each metric's declaration and run **as the viewer** – definable and previewable on a *personal* Gold (promotion only to share and to register the dashboard cube), each emitting a portable MetricFlow-style semantic declaration, honestly *unavailable* (never fabricated) when the lakehouse is unreachable – while **Cube stays on the read path for dashboards only** (dual-run; dashboard migration is Phase 2). **Dashboards** render **natively in the OS** – Apache ECharts on the governed

Cube layer, every panel queried **as the viewer** under per-user row-level security – with Power BI / Tableau / Superset-console bridges over the Cube SQL API. A developer **sos CLI** (Phase 0) brings the same governed door to your own terminal. The OS UI is v1.0: every sidebar tab is a real, brand-themed surface with light/dark theming.

Connections federate the outside world through one governed door: the tab lists connections (All/My/Domain/Company, with app-generated MCP connections folded in), a **vendor-stack-grouped, searchable Supported Connectors** gallery (Microsoft • Google • AWS • Databricks • Snowflake • Salesforce • Kajabi • Atlassian • Open source • Other), a **Custom Connector** for your own REST/GraphQL API or MCP server (one action creates the connection *and* files the egress request), and **Talk to Connectors**. The catalogue spans **operational databases** (PostgreSQL • MySQL • SQL Server • MongoDB via Trino), **code & DevOps** (GitHub), **docs & knowledge** (Notion • Atlassian), **Supabase, messaging & calendar** (Slack • Gmail • Google Calendar • Outlook • Teams – reads auto, sending approval-gated and never automatic), **cloud governance / ML** (Microsoft Entra • Purview • Azure AI Foundry • AWS SageMaker, read-only), data-ingest (Google Drive / OneDrive), the medallion **layer choice** on agent data grants, an admin-enabled **external-warehouse connector** (federate AWS Glue/Athena • Snowflake • BigQuery • Databricks/Delta, plus experimental Fabric/OneLake, through Trino – discover → register → import → **scheduled incremental sync** (full-refresh / append / merge as the dataset's owner, cursor + lookback, presets from every 15 minutes to weekly, auto-pause after repeated failures), no YAML – with PostgreSQL • MySQL • SQL Server surfaced as **sync-capable operational databases**, **Apache Kafka** as a streaming source (configured topics federate as governed tables; an append-only **per-partition offset cursor** lands messages in the lakehouse – no one-time import of an unbounded stream), and **Salesforce** as an API-based operational source (no Trino connector exists, so the sync pulls **SystemModstamp** slices over the REST API page-by-page and streams them into the lakehouse; deletes are not detected in v1 and every pull consumes API quota), and **Kajabi** as its SaaS peer over the public API (OAuth client-credentials from Settings → Public API; JSON:API pages stream into the lakehouse with honest per-resource cursors – purchases by **updated_at**, contacts/customers/orders by **created_at** new-records-only, the rest full-refresh only; deletes never detected)), **one-click Power BI** (a **.pbids** file into Cube's Postgres-wire SQL API, DirectQuery, as the per-domain **bi_<domain>** principal so per-domain RLS re-runs on every query – no embedded password), an **Apache Airflow** connector (governed **trigger_dag** /monitor), and **OpenMetadata** (read/discover of a customer's existing catalog as a Connection, DLS-clamped and **active-only** – archiving an OS dataset soft-deletes its catalog entity, unarchiving restores it). Secrets are write-only; setting up each connector's OAuth app / tokens is the operator's step (every connector ships an install guide). External connectors are off by default and validated against a live source with your own credentials.

Shipped as explicitly-labeled Phase-1 slices (their next phases need new infra or your cloud credentials): Science's guided-train + real training runtime, OpenMetadata scoped write-back, and true per-viewer Power-BI RLS (today's Power BI path enforces per-*domain* RLS via the **bi_<domain>** principal). The developer **sos CLI** is Phase 0 (login • whoami • datasets • query); its roadmap (typed REST, device-code auth, signed-binary distribution, git-through-policy) is in **cli/sos/ROADMAP.md**. The full, versioned history is in **CHANGELOG.md**.

Sovereign Agentic OS – built from permissively-licensed open source for EU data residency. The core is Apache-2.0; bundled components keep their own licenses. This guide is generated from the repository; to update it, edit `docs/Sovereign-Agentic-OS-Guide.md` and run `scripts/build-docs.sh`.